WIDE AREA NETWORKS

Networking Assignment

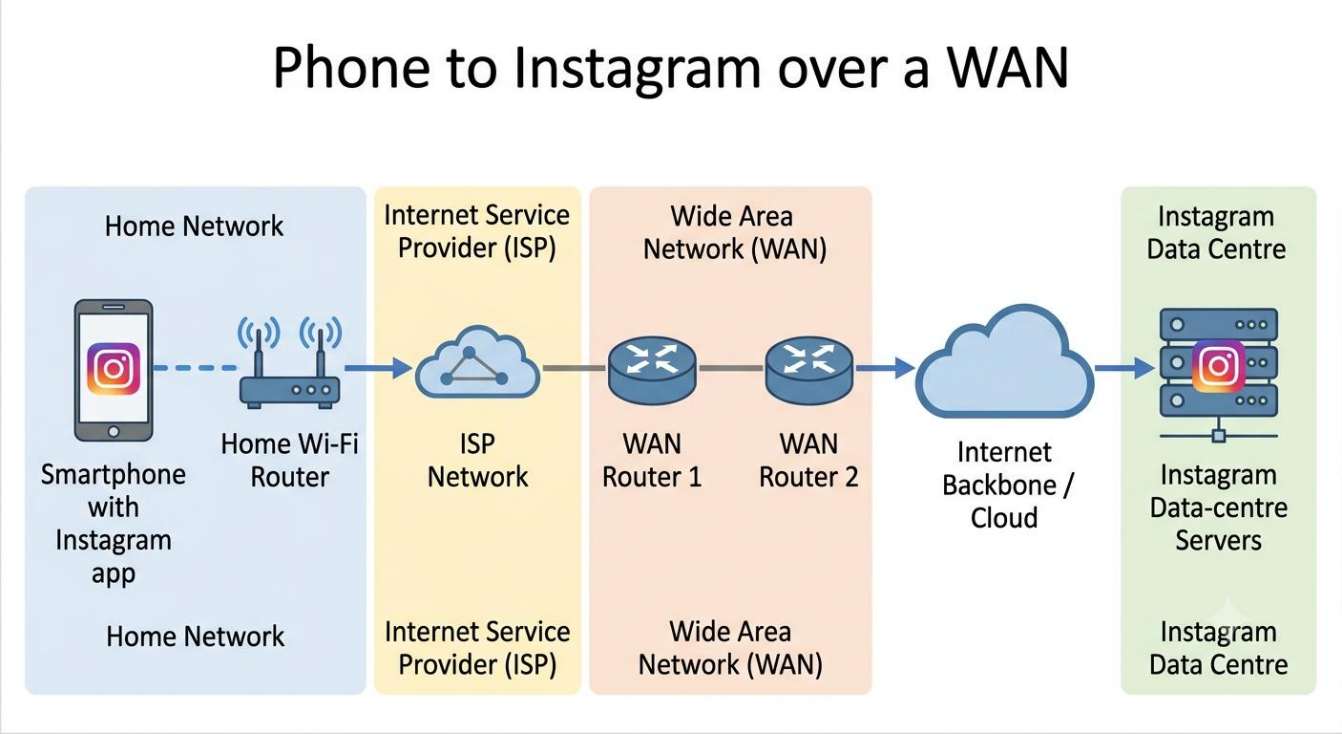
Topic: WAN technologies, VPNs, and business connectivity

Task 1: Path from a Home Wi-Fi Phone to Instagram Servers

When I open Instagram on my phone at home, the phone sends data through the home Wi-Fi router. The router forwards the traffic to my Internet Service Provider (ISP). From there, the data travels through the ISP's network and other WAN routers before reaching Instagram's data centre or cloud servers. The response returns through the network to my phone.

Network Path

PHONE (Instagram App)
↓ Wi-Fi
HOME WI-FI ROUTER
↓
ISP ACCESS NETWORK / ISP ROUTER
↓
WAN ROUTER 1 → WAN ROUTER 2
↓
INTERNET BACKBONE / TRANSIT NETWORK
↓
INSTAGRAM DATA CENTRE / CLOUD SERVERS



Task 2: Comparison of Three WAN Technologies

WAN (Wide Area Network) connects computers and networks across large geographical areas, such as different cities or countries. The following table compares three common ways to connect sites.

WAN Technology	Speed / Performance	Cost	Real-life Use
MPLS (Multiprotocol Label Switching)	Predictable performance; supports traffic prioritisation. Speed depends on the service plan.	Usually high; often priced as a managed business service.	Large companies connecting offices, warehouses, and branches in different cities.
Leased Line	Dedicated link with consistent bandwidth and low variation in performance.	Usually high, especially over long distances.	Banks, data centres, or offices that need a dedicated connection between sites.
Broadband Internet	Can provide high speeds, but performance may vary with congestion and the plan.	Usually low to medium; widely available.	Homes, small offices, shops, and small businesses using cloud apps.

Note: Actual speed and cost depend on the provider, location, bandwidth, service-level agreement, and installation requirements. MPLS is a WAN service technology; a leased line is a dedicated connection, while broadband is a general Internet access service.

Task 3: How a VPN Uses a WAN on Public Wi-Fi

A VPN (Virtual Private Network) creates an encrypted connection, often called a private tunnel, between my device and a VPN server. At a coffee shop, my phone first connects to the public Wi-Fi. When I turn on a VPN, the VPN software encrypts my network traffic and sends it through the public Internet WAN to the VPN server. The VPN server decrypts the traffic and forwards requests to services such as YouTube or Spotify. Replies return through the VPN tunnel to my device.

The encryption helps protect data from people on the same public Wi-Fi network, and the tunnel makes the traffic harder to read while it travels across the WAN. A VPN does not make a person completely anonymous and does not protect against every threat, such as phishing or malware. I should still use HTTPS websites, keep apps updated, and avoid suspicious links.

Image prompt (optional visual to add): Create a simple educational diagram titled “VPN Private Tunnel over Public Wi-Fi”. Show a laptop or smartphone connected to a coffee shop Wi-Fi router, then a public Internet/WAN cloud, then a VPN server, and finally YouTube or Spotify servers. Draw a clearly highlighted encrypted tunnel from the device through the WAN to the VPN server. Use arrows for traffic direction and short labels. Clean blue-and-grey flat vector style, white background, readable labels, no decorative clutter.

Task 4: Recommended WAN for Flipkart Warehouses

For Flipkart warehouses in Ahmedabad, Mumbai, and Delhi, I would recommend an MPLS WAN for the main business network, with a secure broadband VPN connection as a possible backup where appropriate.

Reasons:

1. Reliable communication between locations: MPLS can provide managed connectivity and predictable performance, which is useful for warehouse systems, inventory updates, order processing, and internal applications.
2. Traffic prioritisation and central management: The company can prioritise important business traffic and manage connections between the three cities through a service provider.

The final choice should depend on the required bandwidth, service-level agreement, provider coverage, security needs, and total cost. If keeping costs low is the main priority, site-to-site VPN over business broadband may be a practical alternative.

Conclusion

WANs allow devices and organisations to communicate over long distances. Broadband is suitable for many homes and small businesses, while MPLS and leased lines can support business sites that need managed or dedicated connectivity. A VPN adds encryption to traffic travelling over a public WAN, and a suitable WAN design helps companies connect warehouses and offices across cities.